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RESEARCH ARTICLE

Advancing Breast Cancer Detection: SE-Conformer Framework for Malignancy Detection in Histopathology Images

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ABSTRACT Globally, breast cancer is the second most lethal form of cancer among women, and has high rates of incidence and mortality. Early detection is crucial in improving survival rates. As deep learning technologies continue to advance, they are increasingly being leveraged to automate pathological image classification, primarily through conventional models. Traditional models excel at processing images in which objects are clearly defined against their backgrounds, but they struggle with pathological images because the boundaries between tumor lesions and surrounding tissues are often indistinct. Hence, models that can learn the minute details in these kinds of images are very essential.We propose a hybrid approach that integrates convolutional networks with attention mechanisms to achieve high classification performance while reducing computational complexity. To enhance feature representation in convolutional neural networks, we introduced an improved convolutional block, termed the SE-Res-Conv Block, which incorporates Squeeze-and-Excitation (SE) attention mechanisms within a residual convolutional framework. The extracted features are then processed by a Conformer block, which further refines them by emphasizing the most relevant regions in the input feature map. Finally, a dense output layer is used for classification. The proposed model is evaluated on both the BACH dataset and all four magnification levels of the BreakHis dataset. Experimental results, obtained through K-fold cross-validation, indicate that the model performs optimally on 200× magnification images, achieving an average accuracy, precision, and recall of 0.97. These results underscore the model's effectiveness and its potential applicability in medical image classification tasks.

INDEX TERMS Histopathology images, breast cancer, BreakHis, squeeze and excitation block, convolution, attention mechanism, conformer.

I. INTRODUCTION

One of the diseases that is affecting the women population in the world is breast cancer.It is becoming one of the most common cancers diagnosed in women.Recent surveys showcase that the breast cancer incidence rates are increasing gradually every year.Hence proper diagnosis is very essential for the timely treatment of the patient to lower the mortality rate due to this disease.Early detection and accurate diagnosis significantly increase survival rates and also help eliminate the chances of recurrence.In [1] authors have done a study on the prediction of recurrence,highlighting the need for advanced diagnostic methods and technologies. Despite significant advancements in screening techniques and treatment modalities, breast cancer remains a formidable challenge in clinical practice due to its heterogeneous nature, complex pathology and diverse molecular subtypes. The heterogeneity of tumour morphology, genetic alterations and receptor status between patients makes precise diagnosis and individualized treatment planning challenging. Moreover, problems like early detection in dense breast tissue, differentiation between benign and malignant lesions, treatment response prediction and reduction of false positives and negatives in screening programs remain to impede

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efficient disease management. These issues underscore the imperative for more precise and reliable diagnostic tools to improve patient outcomes. To assist oncologists in the effective identification of cancerous areas in different types of cancer, researchers have been trying various imaging methods employing samples of data taken from patients. There are some imaging methods used for the diagnosis of breast cancer at different stages. Currently, mammography is considered to be the gold standard for early screening due to its non-invasive and cost-effective nature in identifying possible abnormalities. Ultrasound imaging often serves as a complementary technique to mammography, especially when dense breast tissue reduces mammographic effectiveness. Furthermore, Magnetic Resonance Imaging (MRI) is frequently utilized for screening high-risk patients and is essential for pre-surgical planning. Histopathology imaging plays a pivotal role in definitive diagnosis by enabling microscopic examination of excised or biopsied tissue samples. Even-though each of these modalities serves a unique purpose, contributing to comprehensive cancer diagnosis and treatment planning, histopathology images are used by pathologists due to its significance in detecting the exact cancerous regions from the images. This is an invasive method in which the tissue samples of the patients are taken and then the digitized microscopic image is saved after applying proper stain on the tissue for better visibility of various cellular components in it. The characteristic of the cellular structure is then analyzed to detect the abnormalities in the cell shape, size, tissue patterns, nuclei shapes etc which helps in detecting benign and malignant regions. Histopathology images are captured by processing biopsy specimens through various steps including staining, which is very essential for distinguishing various cellular structures in the specimen. Haematoxylin and Eosin (H&E) staining is the most commonly used staining technique, where the nucleus in the cells turns blue on applying haematoxylin stain and eosin stains cytoplasm and remaining structures to pink, providing contrast necessary for cellular analysis. These stained tissue sections are then digitized using high-resolution whole-slide scanners at varying magnification levels, to capture detailed morphological features. The digitized histopathology slides enable pathologists and computational algorithms to analyse tissue architecture and cellular structures for diagnostic purposes.

As the Histopathology images are captured at different magnification levels, the information captured in each of the images obtained at different magnification varies compared to the other and hence various CAD techniques have been proposed in this domain for more accurate prediction of cancer and the possible type of cancer in the sample images. There are various datasets available which helps in modelling different computer aided diagnosis systems. One such publicly available dataset is BreakHis [2],having images captured at different magnification levels 40x,100x,200x,400x, each level of magnification reveals information at different levels. The images in this dataset are marked as either benign or malignant and further into 4 subclasses per category representing various cellular and granular features based on the type of cancerous or non-cancerous regions. Another widely used dataset is the Breast Cancer Histopathological Database (BACH) [3], containing annotated H&E-stained whole-slide images with four different categories: normal, invasive carcinoma, benign and in situ carcinoma.

Extensive research has been conducted for breast cancer detection using histopathology images, encompassing various approaches aimed at improving diagnostic accuracy and efficiency. These approaches involves traditional feature extraction with machine learning methods, deep learning-based techniques, and various hybrid models. Traditional methods often rely on handcrafted feature extraction using various image processing techniques and then use of various machine learning classical classifiers. In contrast, deep learning methods, especially Convolutional Neural Network (CNNs) based models, have demonstrated superior performance with the ability in learning hierarchical features automatically from raw images. Recent studies have also explored transfer learning, attention mechanisms, and ensemble models to enhance classification, segmentation, and grading of breast cancer histopathology images. Furthermore, explainable AI (XAI) techniques [4] are gaining attention to improve the interpretability of automated diagnostic systems, making them more trustworthy for clinical use.

The following part of this paper is presented as follows: A summary of related work is given in Section II, Section III presents the main contributions of our work, Section IV explains the datasets used and the detailed explanation of the proposed architecture. Experimental results including ablation studies are included in section V. Finally, we conclude by presenting the summary of our work in Section VI.

II. RELATED WORKS

Numerous research studies have explored deep learning techniques for classifying breast cancer using histopathology images. In the comparative study [5], the authors evaluated various deep learning models applied to breast cancer classification using histopathological images. The primary focus of the study is on assessing the performance of several Convolutional Neural Network (CNN)-based models, including popular pretrained architectures such as VGG, ResNet, Inception, and DenseNet. These models were fine-tuned through transfer learning techniques to improve classification accuracy. The study shows how deep learning can automatically learn important features from raw histopathology images, eliminating the need for manual feature selection. It also offers a thorough comparison of these models, evaluating their accuracy, precision, recall, and efficiency across various datasets and magnification levels. Benhammou et al. [6] conducted an extensive survey on deep learning approaches applied to the BreakHis dataset, a widely recognized benchmark for classifying histopathological breast cancer images. The study systematically reviews and categorizes various deep learning architectures

and techniques designed to differentiate between benign and malignant breast tumors using BreakHis images captured at different magnification levels (200x, 400x, 40x, and 100x). Saini and Susan [7] proposed DCGAN-based data augmentation approach to address the prevalent problem of class imbalance in breast cancer histopathology datasets. In their study, they introduced an approach that combines deep transfer learning with focused data augmentation. Class imbalance can result in biased models with poor generalization and reduced sensitivity to minority classes. To address this challenge, the authors utilize pretrained deep Convolutional Neural Networks (CNNs), including VGG and ResNet, through transfer learning, allowing for effective feature extraction even with limited training data.Methods such as geometric transformations, color jittering, and synthetic image generation are generally applied to increase data diversity and promote a more balanced class distribution.

The authors present GC-EnC [8], an ensemble approach that combines predictions from multiple Convolutional Neural Networks (CNNs) using a copula-based dependency model. This technique captures complex interdependencies among the CNN outputs more effectively than traditional averaging or voting methods. While GC-EnC leverages the strengths of diverse CNN architectures to enhance predictive performance, it also introduces higher computational demands, including increased training time, memory usage, and slower inference speed compared to individual CNN models or simpler ensemble strategies.

Many research works have focussed on different feature fusion mechanisms for improving the classification accuracy. Das et al. [9] presents a generalization framework that enhances the robustness and reliability of breast cancer detection systems in histopathological images across different domains, without need for labelled data from new domains. Their model got an accuracy of 0.88 on 200x magnification images in BreakHis dataset.

Wang et al. [10] introduced a novel method called FE-BkCapsNet for classifying breast cancer histopathological images. This method leverages the strengths of both Convolutional Neural Networks (CNNs) and Capsule Networks (CapsNets) to enhance classification performance. Their approach involved a dual-channel structure that simultaneously extracts convolutional and capsule features, which integrates spatial and semantic features to generate new capsules for obtaining more discriminative information. The method was tested on the BreaKHis dataset, achieving classification accuracies across different magnifications with highest accuracy of 0.94 for 100x.BCHisto-Net [11], a model to classify the images of 100x magnification alone uses a feature fusion technique with extraction of global features which represents the abstract of images and the local features focussing on potential ROIs achieving a classification accuracy of 0.89. Another work proposed by Zerouaoui and Idri [12] did a comparison of different hybrid architecture with deep learning models for feature extraction followed by classification with different machine learning classifiers, Densenet101 with MLP classifier gave the highest accuracy rate of 0.93 for 200x magnification images. Similar work proposed in [13] used multiple DL models for feature extraction followed by feature concatenation and then use of fully connected layers for classification.

In [14], He et al. proposed a transformer-based architecture named DecT that combines color deconvolution with a transformer-based deep learning architecture. The model employs color deconvolution to separate stain components, enhancing the visibility of key tissue structures while minimizing color-based noise. It then utilizes a transformer model for feature extraction to enable further classification. The model obtained an average accuracy of 0.93 in BreakHis dataset.

In [15] authors suggested a hybrid framework for the early detection and multi-class classification of breast cancer from histopathological images, combining both deep learning-based features (CNN) and handcrafted features to improve diagnostic accuracy. The deep features and handcrafted features are fused to create a comprehensive feature set. The effectiveness of such hybrid model may depend heavily on the feature selection and dimensionality reduction techniques used after feature fusion. Snigdha and Nair [16] have proposed a similar model considering both hand crafted and deep learning features for classification task using IDC dataset. In [17], the authors focused solely on mitotic features for breast cancer prediction, utilizing both handcrafted features and CNN-derived features for classification.

MDFF-Net [18] proposed by Cheng Xu et.al focus on extracting and fusing features from multiple dimensions to enhance the representation ability of the model. The authors have proposed two separate feature extraction networks, one for extracting morphological features and the other one based on mechanisms to capture spatial features effectively. Finally a third feature fusion network classifies the images. The paper [19] presents a combined Feature-Vector Based (MIL-CNN) network for classifying breast cancer using histopathological images. This method leverages a Multiple Instance Learning framework, where whole-slide images are divided into smaller patches (instances), and only bag-level labels are required, reducing the need for detailed annotations. Each patch is processed to extract deep features using CNNs, which are then combined with handcrafted features such as morphological descriptors and texture features to form a comprehensive feature vector. The quality of classification heavily depends on effective patch selection strategies. Some of the works [20], [21], [22], [23], [24] focused on transfer learning models. The approach in [20] is to handle magnification-dependent classification by training and testing on images at specific magnification levels, while magnification-independent classification is achieved by incorporating multi-scale images during training to ensure robustness across varying magnifications. In [25]

authors have proposed an approach which is a hybrid of two Deep Learning models, Inspection-ResNetv2 and EfficientNetV2-S using pretrained weights from ImageNet.

Reference [21] proposes an Efficient Breast Cancer Classification Network that incorporates a Dual Squeeze and Excitation (DSE) mechanism to enhance feature representation in histopathological images. The method integrates two types of attention mechanisms—channel attention and spatial attention, to recalibrate and emphasize the most informative features in both the channel and spatial dimensions. However, it fails to accurately classify certain samples affected by imaging artifacts, such as blurriness and poorly defined cell boundaries.

In [26], the authors utilized pretrained models for a single magnification image and applied them to classify images at other magnifications, aiming to improve accuracy. The method addresses the challenges posed by variations in image domains and resolutions commonly found in histopathological data. The model employs domain adaptation techniques to reduce distribution discrepancies between datasets from different sources and integrates multi-resolution learning, allowing it to process and learn from images at varying magnifications. Even though, the model achieves improved generalization it has some limitations, including increased computational complexity due to handling domain and resolution variations simultaneously.

Since the images have certain relevant areas that contribute in identifying the malignancy, many works have used attention-based mechanisms for better contextual learning. DRDA-Net [27] proposed by Soham et al, introduces a dual-shuffle attention network based on channel attention mechanism and obtained the highest classification accuracy of 0.98 for 400x images of BreakHis dataset. In [28], the authors employed the VGG19 model for image feature extraction, followed by processing through a CBAM block. The output was further passed through an additional feature extraction block before classification. Using pretrained CNN models for feature extraction followed by use of two sequential CBAM was the approach used by Ashurov et al. in [29]. Krishna et al. [30] in their work integrated an attention branch into the DarkNet19 network for the classification of BreakHis images under each magnification.

Vanitha et al. in [31] utilized a Transformer model with an external attention mechanism. A CNN was employed to extract local features, which were then processed by External Attention Transformer (EAT) modules to capture global dependencies. Finally, an attention-based fusion module integrated these multi-scale features for improved classification.

Yan et al. [32] proposed a model with two networks named PCSAM-Net and ResCBAM-Net, the first one to extract features patch wise and the second one to extract features at a global level and then classification was done using the fused features from both networks. Another attention model named FCCS-Net proposed [33] by Maurya et al. advocates the integration of attention mechanisms inside the base model at various levels achieving the highest classification accuracy for 40x magnification compared to other three in the BreakHis dataset.

Hao et al. in [34] proposed a two-stage classification model utilizing the Swin Transformer for global feature extraction in the first stage. The extracted features were fused with those obtained from the second stage, where a Heatmap Cropping module was employed to extract specific lesion regions. These refined features were then fed into a second Swin Transformer block, achieving an accuracy of approximately 0.97 across different magnifications. Several works have been proposed in literature [35], [36], [37] that use Vision Transformers for breast cancer classification in histopathology images. Recently many hybrid models combining convolution with transformers have been proposed. In [38] authors have proposed a hybrid model with EfficientNetV2 and Vision Transformer and in [39] a custom CNN model is used for initial feature extraction, followed by a Transformer block for capturing global dependencies. The extracted features are then fused with the output features from an Xception network to enhance representation learning for improved classification performance. Mahmood et al. [40] introduced a novel convolution transformer-based architecture using adaptive convolution blocks for feature extraction followed by Dynamic attention blocks for further refined feature extractions. Their model performed well for different histopathology datasets but gave comparatively less result for BreakHis dataset. In [41], authors have proposed a model named SwinCNN which integrates the feature extraction capabilities of convolution layers and further the SwinTransformer blocks with a patch merging block in between.This method achieved an accuracy of around 0.98 for binary classification task using BreakHis dataset.

Despite extensive research in this field, balancing computational cost and efficiency continues to be a major challenge. Therefore, developing lightweight models that maintain high performance is of paramount importance.

In this study, we introduce a novel convolution-attention-based architecture incorporating squeeze-and-excitation (SE) blocks within residual convolutional units, named SE-Res-Conv blocks to enhance channel-wise feature extraction of most relevant features, thereby reducing computational complexity. This is followed by a conformer block for further refinement of the feature map prior to classification. The proposed model's performance is evaluated on the BreakHis and BACH datasets for binary and multi-class classification tasks.

III. PROBLEM ADDRESSED

In this work, we present a novel convolution-attention architecture designed for binary and multiclass classification tasks on the BreakHis and BACH image datasets. The novelty of our model stems from its unique feature extraction approach, which incorporates three SE-Residual Convolution Blocks—a custom architecture that integrates

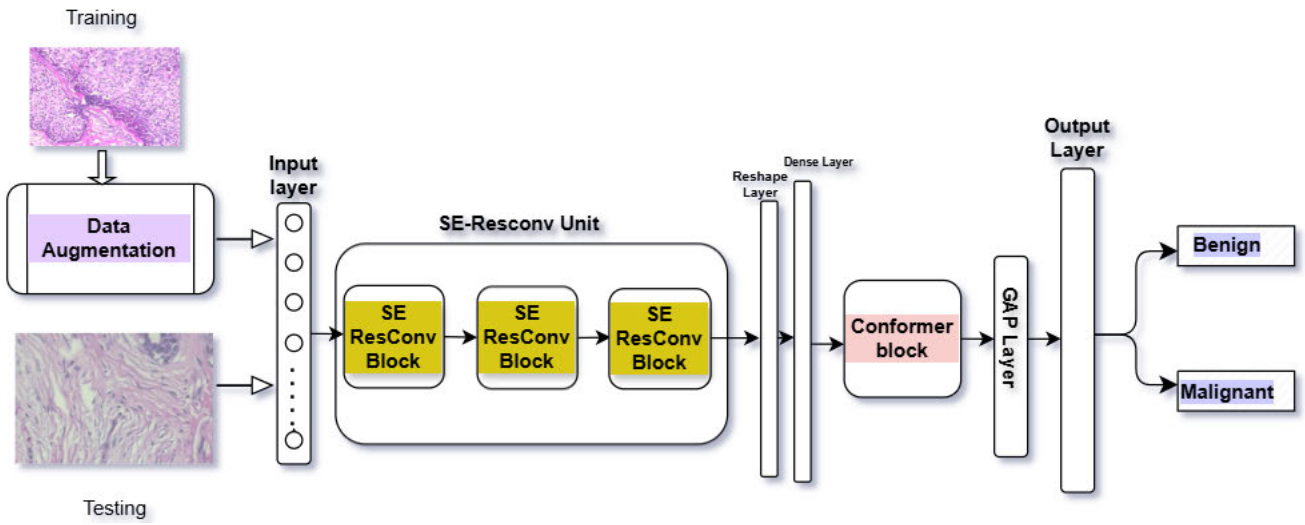

FIGURE 1. SE-Conformer framework.

squeeze-and-excitation (SE) mechanisms into residual convolutional blocks. This design enhances channel-wise feature recalibration and significantly improves the model's representational capacity. This SE-Residual-Convolution Unit with three SE-Res-Conv blocks in it forms the first stage of our architecture and serves to extract rich, hierarchical features from input images. Building upon this, the second stage incorporates a modified Conformer block [42], which synergistically combines convolutional operations and self-attention mechanisms to effectively capture both local and global dependencies in the learned features. Together, these two components enable our architecture to achieve improved classification performance on histopathological image datasets. SE-Res-Conv Block integrates Squeeze-and-Excitation (SE) attention mechanisms [43] within a residual convolution framework to improve feature representation in convolution neural networks. The proposed block combines standard convolution, batch normalization, and non-linear activation with channel-wise attention provided by the SE module, which dynamically adjusts feature responses by capturing interdependencies between channels. The output from the SE-Res-Conv Block is then fed to a conformer block that combines the strengths of convolutional neural networks and attention mechanism to capture both local and global dependencies in the feature map generated by the SE-Res-Conv blocks.

The main contributions of this work are summarized as follows:

- Incorporation of a custom channel-attention based feature extraction block termed as SE-Res-Conv block
- A novel, computationally efficient architecture with three SE-Res-Conv Blocks based on channel attention to generate feature map followed by a conformer block for capturing local and global dependencies in the feature map for classification.

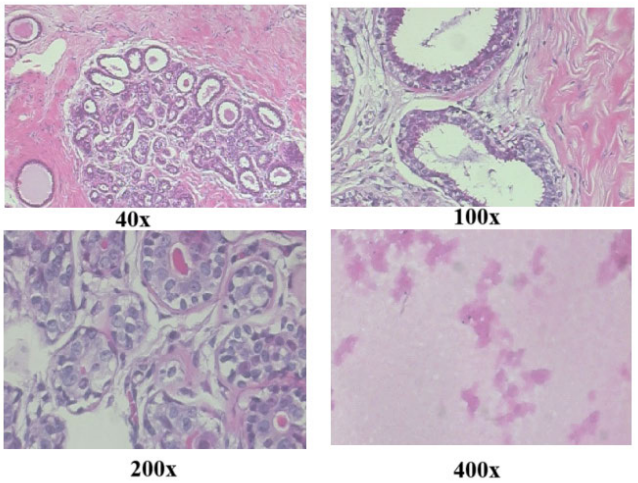

FIGURE 2. BreakHis sample images.

- Model performance tested on Magnification-independent and magnification-dependent classification tasks for binary and multi-class classification with individual and combined datasets.

IV. METHODS AND MATERIALS

A. DATASET

We have used two publicly available datasets for our experiments: BreakHis and BACH dataset.

1) BREAKHIS

BreakHis dataset comprising 7,909 microscopic images of resolution 700 × 460 in .png format. There are four categories of images obtained at different magnification levels (40 ×, 100 × , 200 × , and 400 ×) under two classes, benign and malignant, with 2480 samples under benign

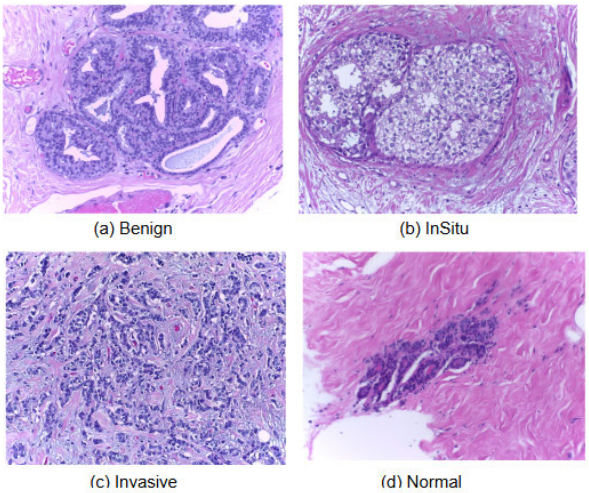

FIGURE 3. BACH sample images.

TABLE 1. Image distribution across different magnifications.

Magnification	Benign	Malignant	Total
40x	625	1370	1995
100x	644	1437	2081
200x	623	1390	2013
400x	588	1232	1820
Total	**2480**	**5429**	**7909**

category and 5429 samples under malignant category as shown in Table 1. Each of these classes are further subdivided into four subclasses as follows: adenosis (A), fibroadenoma (F), phyllodes adenoma (PT), and tubular adenoma (TA) in benign category, whereas ductal carcinoma (DC), lobular carcinoma (LC), mucinous carcinoma (MC), and papillary carcinoma (PC) are the four subclasses in malignant category. Figure 2 depicts sample images at different magnification. For our experiment we have used images of all the four magnifications as shown in Table 1.

2) BACH

The performance of proposed model is demonstrated using another publicly available dataset named the BACH dataset, which contains 400 high-resolution (2048 × 1536 pixels) breast histology microscopy images divided into four classes: normal, benign, in-situ carcinoma, and invasive carcinoma with each class having equal number of images in it. figure 3 depicts sample images from each category.

B. PROPOSED METHODOLOGY

The architecture diagram of our proposed approach is illustrated in figure 1, which comprises three Squeeze-and-Excitation-Residual-convolution blocks (SE-Res-Conv Block) and a Conformer block. The SE-Res-Conv Block is an intrinsic feature extraction module which harnesses the power of convolution, residual connections and SE blocks for suppressing irrelevant information and passing only the relevant data with certain weightage for further processing. The initial feature map generated from the SE-Res-Conv Block is then fed to another block called Conformer Block, which is nothing but a fusion of convolution-attention mechanism as a sandwich between two feed forward layers. This block unifies the power of convolution layer and transformer's self-attention mechanism for capturing local dependency and global dependencies respectively.

As the prediction of cancerous regions from histopathology images is a highly challenging task, it is very important to have a model that can understand the variability and similarities of each region in an image, hence applying a custom model incorporating the strength of convolutional neural networks (CNNs) and attention mechanism will help in capturing both local and global dependencies within the image very effectively. Our proposed architecture is designed with a series of interconnected layers which is capable of reshaping the input images, extract only the relevant features from it and based on that classify the images into different classes as benign and malignant and also to subclasses with good accuracy.

The architecture begins with an input layer configured to receive images of dimensions 128 × 128 × 3, followed by three sequential Squeeze-and-Excitation Residual Convolution (SE-ResConv) blocks, the output from the third SE-ResConv block is reshaped before passing to a fully connected layer which connects the SE-ResConv Unit and conformer block. The output from the SE-Res-Conv unit is transformed into a 128-dimensional feature vector via a fully connected layer, which is then fed into a customized Conformer block,which comprises a Multi-head self-attention unit followed by a convolution unit embedded between two fully connected layers. Each of the unit is wrapped with residual connections for stable gradient flow. Global average pooling is applied to the output generated from the Conformer block with a dropout of 0.5 and then fed to the output layer for classification.

The proposed model has been employed for both binary and multi-class classification tasks on the BreakHis and BACH datasets, considering both magnification-independent and magnification-dependent approaches. The following subsections explains the design of SE-Res-Conv Block and the Conformer block in detail.

1) INPUT LAYER

This layer reshapes the input image to dimensions of $128 \times 128 \times 3$, ensuring compatibility with the first SE residual convolutional block according to the defined parameters.

2) SE-RES-CONV BLOCK

As depicted in the figure 1,the input image is reshaped and then passed through a Squeeze and excitation residual

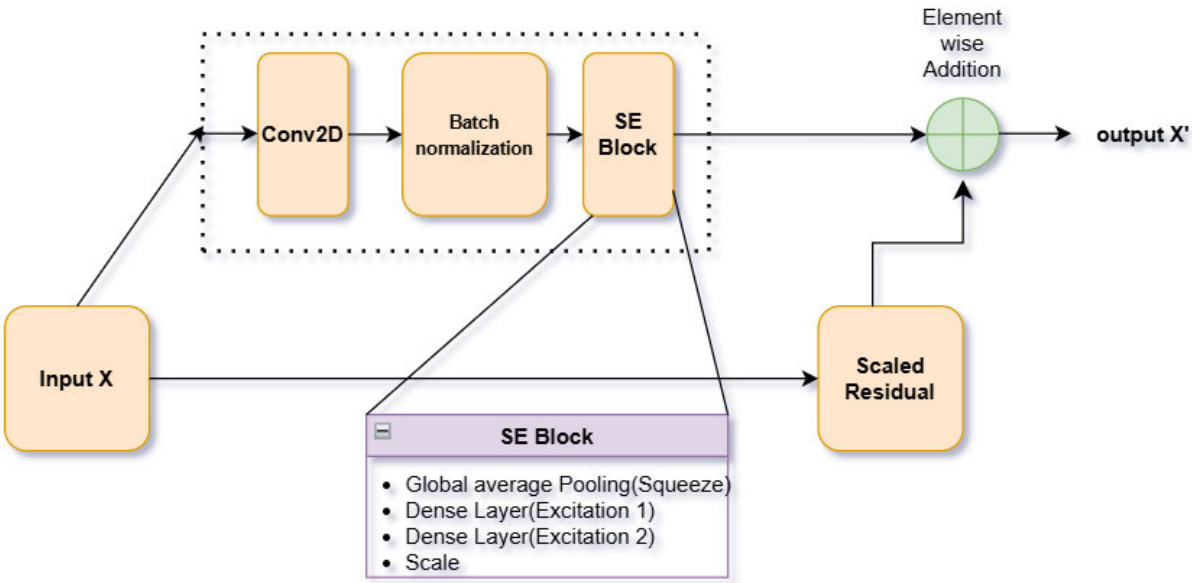

FIGURE 4. SE-residual convolutional block.

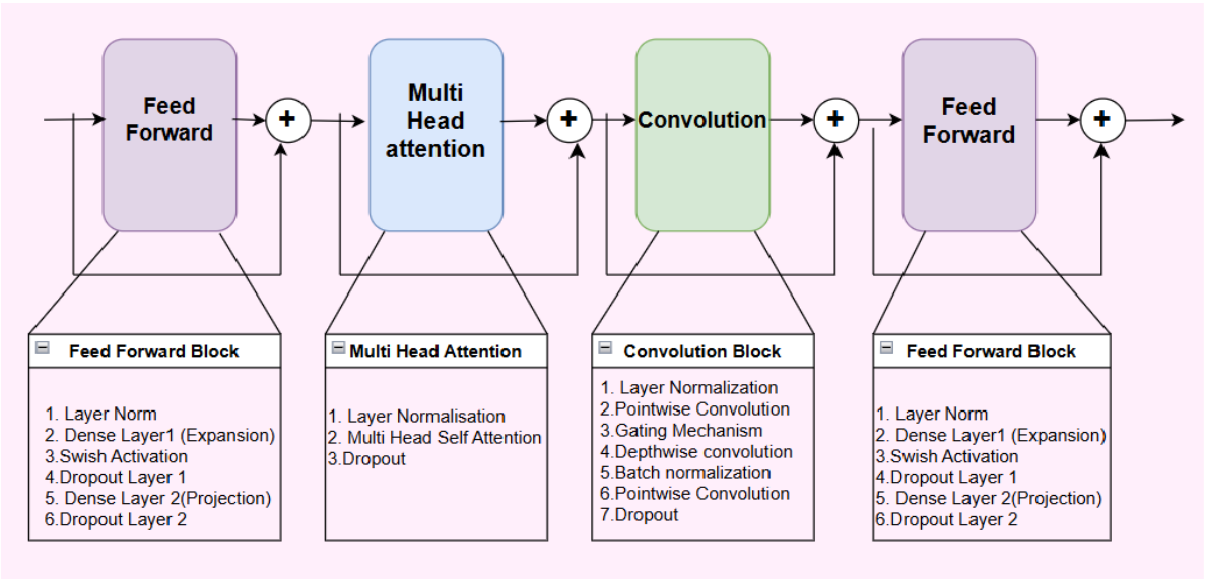

FIGURE 5. Conformer block.

convolution Unit having three SE-res-conv block in it. A SE-res-conv block is an integration of a Squeeze-and-Excitation (SE) block into a Residual Convolution Block, as shown in figure 4. The Squeeze-and-Excitation (SE) block, first introduced in the SENet architecture, refines feature representation by modeling interdependencies among channels, allowing the network to focus on the most relevant features. It begins with a squeeze operation, utilizing Global Average Pooling (GAP) to condense spatial information across the entire feature map, effectively reducing each channel's spatial dimensions to a single representative value. GAP ensures that each channel's importance is determined based on its global context rather than local activations alone. The pooled features are then passed through an excitation step which involves two fully connected layers. In the first fully connected bottleneck layer, the number of neurons is reduced by a factor of 16(reduction ratio),enforcing channel-wise compression. This reduction is done to limit the computational complexity and support selection of meaningful features.

ReLU activation is employed to introduce non-linearity, followed by a second fully connected layer with a sigmoid activation function that expands the compressed representation back to the original channel dimension. This process

yields a set of per-channel scaling factors, capturing the relative importance of each channel.

In the final scaling step, the computed weights are multiplied with the original feature map. These scaling factors, constrained between 0 and 1, function as attention weights, enabling the network to emphasize the most critical features. This operation enhances the representational capacity of the network by adaptively reweighting channel-wise features, allowing it to focus on the most informative aspects of the input while suppressing less relevant information.

Algorithm 1 SE-Conformer forward propagation

Input: Training dataset $\{(X_i, Y_i)\}_{i=1}^{N}$, initialization parameters,number of epochs
Output: Trained model
Initialize parameters of SE Residual blocks and Conformer block;
for *epoch* $= 1$ **to** *num_epochs* **do**
  foreach (X_i, Y_i) *in training dataset* **do**
    - Extract features using SE residual convolution blocks: $F_{\text{conv}} \leftarrow \text{SE_Residual_Conv}(X_i)$;
    - Reshape feature maps for sequential modeling: $F_{\text{seq}} \leftarrow \text{Reshape}(F_{\text{conv}})$;
    - Process with feed-forward module: $F_{\text{ff1}} \leftarrow \text{FeedForward}(F_{\text{seq}})$;
    - Apply multi-head self-attention module: $F_{\text{attn}} \leftarrow \text{MultiHeadSelfAttention}(F_{\text{ff1}})$;
    - Apply convolution module for local feature modeling: $F_{\text{attn}} \leftarrow \text{ConvolutionModule}(F_{\text{attn}})$;
    - Process with another feed-forward module: $F_{\text{ff2}} \leftarrow \text{FeedForward}(F_{\text{conv_mod}})$;
    - Compute global representation: $F_{\text{global}} \leftarrow \text{GlobalAveragePooling}(F_{\text{ff2}})$;
    - Compute loss: $\mathcal{L} \leftarrow \text{Loss}(F_{\text{global}}, Y_i)$;
    - Compute gradients: $\nabla\mathcal{L} \leftarrow \text{ComputeGradients}(\mathcal{L})$;
    - Update parameters: $\theta \leftarrow \text{UpdateParameters}(\theta, \nabla\mathcal{L}, \text{learningrate})$;
  end
end

The SE-Res-Conv Block integrates convolutional layers, residual connections, and a squeeze-and-excitation (SE) mechanism to improve feature extraction and overall model efficiency. Initially, the input tensor passes through a convolutional layer configured with specific filters, kernel size, and strides, followed by Batch Normalization (BN) to promote stable training and faster convergence. Subsequently, the SE block is used to re-calibrate and enhance the significance of the output channels. To maintain compatibility with residual connections, if the dimensions of the input tensors differ from the output, a 1×1 convolution is applied to the residual path for alignment. Finally, the recalibrated output and adjusted residual are summed via an add operation, enabling gradient flow through shortcut connections and mitigating vanishing-gradient issues in deep networks. We have incorporated this block for our model as this combination of SE blocks and residual learning ensures that both spatial and channel-wise informations are effectively utilized, which resulted in improved performance for our image classification task with minimal computational overhead.We have experimented the performance by varying the number of blocks in this unit.

3) CONFORMER BLOCK

We used a fusion of convolution and the attention mechanism, which is nothing but a modified version of the Conformer block originally proposed to extract local and global dependencies in sequential data, particularly for speech recognition. We have fine-tuned this structure to make it suitable for our application. This block integrates four key components: two Feed-Forward Modules, a Multi-Head Self-Attention (MHSA) Module, and a Convolution Module, arranged in a sandwich-like structure as shown in figure 5. The Feed-Forward Module applies layer normalization, followed by two dense layers with L2 regularization and a Swish activation, introducing an expansion factor to enhance expressiveness while employing a half-step residual connection for stable learning.

The feed forward module uses layer normalization to stabilize training and subsequently feeds the input through two dense layers with a Swish activation function in between. The first dense layer increases the dimensionality (regulated by an expansion factor), and the second projects it back to the original dimension. Dropout layers are added for regularization, preventing overfitting by randomly setting a proportion of inputs to zero during training. After the first feed-forward module, the output is given to a multi-head self-attention module to enable the model to attend to multiple locations in the input sequence at the same time. By allowing the model to capture relations between different elements in the sequence, this module improves contextual awareness and facilitates improved feature representation.

The convolution module uses point-wise and depth-wise convolutions to further enhance feature extraction. We used a Multiply-Sigmoid Gating Mechanism, so that dynamic selection of important features can be made based on learned gating signals. This allows model performance to be enhanced by picking out significant features while rejecting the noisy ones. Finally, another Feed-Forward module is applied, ensuring a balanced representation of hierarchical features.

Each submodule is interleaved with dropout layers for regularization, maintaining robustness while preventing overfitting. This structured combination of self-attention, gating mechanisms, and convolution efficiently captures both global dependencies and local contextual information in the feature map generated by the SE-Res-Conv block, making it highly effective for the breast cancer classification task.

4) GLOBAL POOLING AND CLASSIFICATION

After processing by the Conformer block, the output is passed through a Global Average Pooling Layer, which aggregates

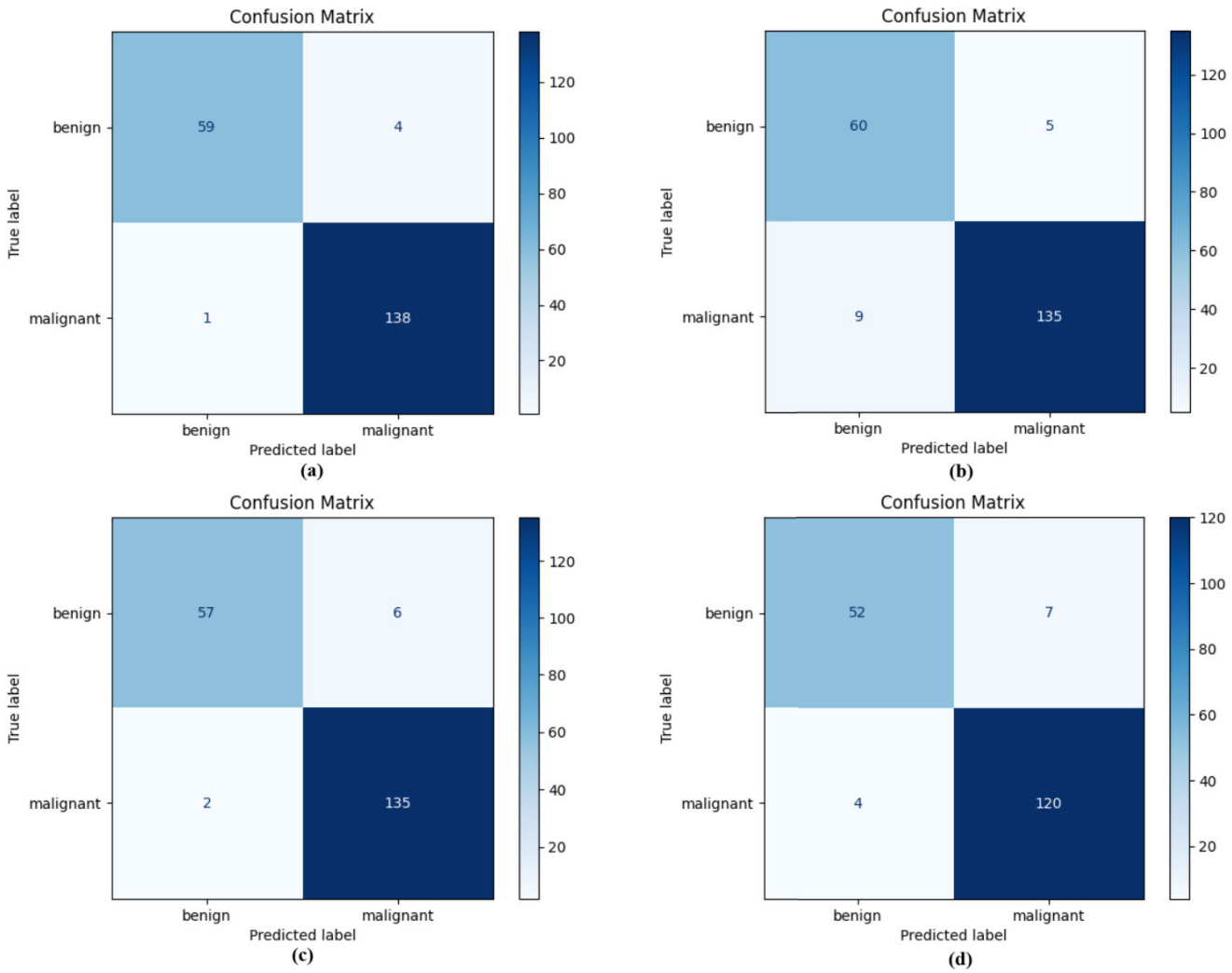

FIGURE 6. Confusion matrix for (a) 200x (b) 100x (c) 40x (d) 400x.

information across all spatial positions in the sequence into a single vector representation. This pooling operation reduces the dimensionality while retaining essential information about feature distributions. Following pooling, a Dropout Layer is applied for additional regularization before reaching the final classification stage. The last component is a Dense Output Layer, which uses a sigmoid activation function to produce binary classification output. This layer generates probabilities indicating class membership for each input image based on its learned features.

V. EXPERIMENTAL RESULTS

The experimental setup was implemented on a system equipped with an AMD Ryzen 7 5800H, NVIDIA RTX 3050 GPU, and 16GB RAM, running on Windows 11 with TensorFlow for model training and evaluation. We conducted a comprehensive performance evaluation of our model using the BreakHis and BACH datasets.

The BreakHis and BACH datasets were used for model training and evaluation. For the experiments, the images were split into training and testing sets using an 80:20 ratio for each of the experiments with different magnification, 10% of the test set was further reserved for validation purposes to monitor model performance during training. The split was performed in a stratified manner to maintain the class distribution in both subsets.The training set was further augmented with strong augmentation for minority class and mild augmentation strategy for majority class. Table 3 shows the number of images used for testing and training from each of the dataset. The images in the training and test set were taken after a random shuffling of the dataset for each of the experiment to avoid bias and overfitting.

The hyperparameters used for model training is shown in Table 2.

Table 5 summarizes the model performance in the binary classification task for both magnification independent and magnification dependent images. The results indicate that

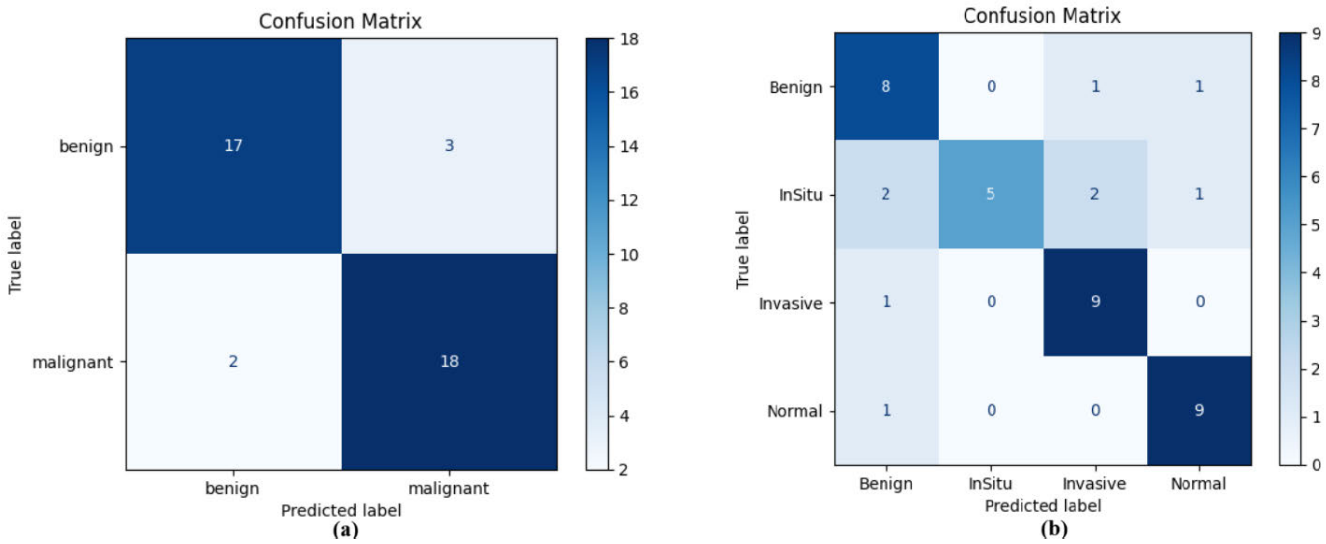

FIGURE 7. Confusion matrix for (a) Binary (b) Multi-class classification in BACH dataset.

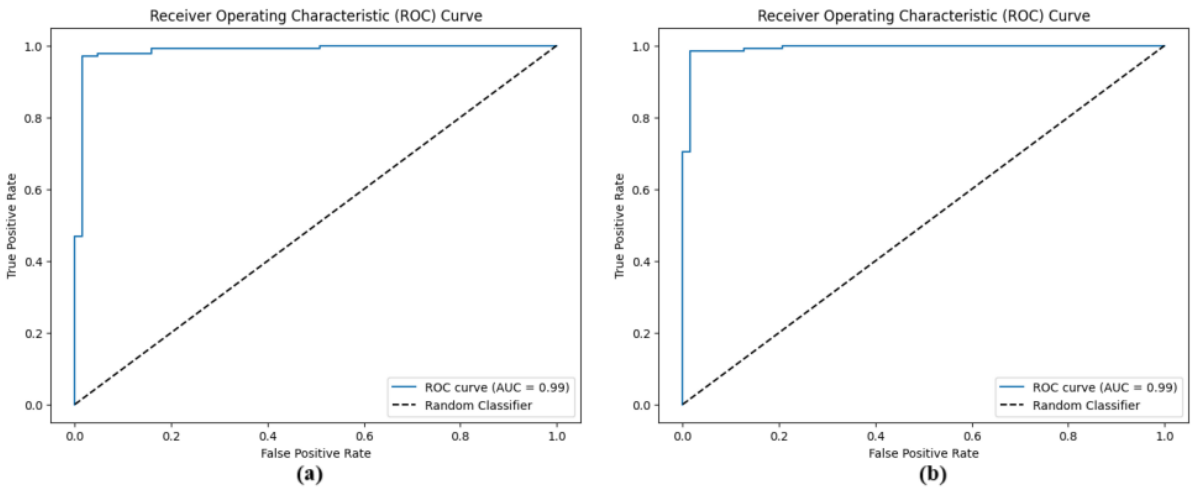

FIGURE 8. ROC curve (a) Binary classification with 200x images from BreakHis (b) Binary classification with 200x BreakHis and BACH images.

TABLE 2. Hyperparameter configuration.

Hyperparameter	Value
Batch Size	32
Epochs	150
Optimizer	Adam
Learning Rate	0.0001
Early Stopping Patience	15
ReduceLROnPlateau Patience	3

TABLE 3. Training and testing distribution across datasets.

Dataset	Training		Testing	
	Malignant	Benign	Malignant	Benign
BreakHis40x	1296	1096	137	63
BreakHis100x	1349	1149	144	65
BreakHis200x	1312	1112	139	63
BreakHis400x	1185	985	124	59
BACH	640	640	20	20

the 200× magnification achieves the highest classification accuracy among all magnifications. The confusion matrices corresponding to each magnification level are presented in Figure 6. Furthermore, experiments were performed for the multiclass classification, with Table 4 providing the

TABLE 4. Classification report-BreakHis.

Class	Precision	Recall	F1-Score	Support
Adenosis	0.77	0.91	0.83	11
Ductal Carcinoma	0.88	0.90	0.89	90
Fibroadenoma	0.81	0.93	0.86	27
Lobular Carcinoma	0.75	0.75	0.75	16
Mucinous Carcinoma	0.78	0.70	0.74	20
Papillary Carcinoma	0.71	0.77	0.74	13
Phyllodes Tumor	1.00	0.36	0.53	11
Tubular Adenoma	0.71	0.71	0.71	14
Accuracy			0.82	202
Macro Avg	0.80	0.75	0.76	202
Weighted Avg	0.83	0.82	0.82	202

TABLE 5. Classification performance of BreakHis dataset.

Magnification	Precision	Recall	F1-Score	Accuracy
40x	0.96	0.99	0.97	.96
100x	0.96	0.94	0.95	.93
200x	0.97	0.99	0.98	.98
400x	0.94	0.97	0.96	0.94
Combined	0.94	0.97	0.96	0.94

detailed classification report. The confusion matrix for binary classification is illustrated in Figure 6, while Figure 10 shows the confusion matrix for multiclass classification.

Similarly, we conducted experiments using the BACH dataset for both binary and multi-class classification tasks. As the number of images in this dataset is significantly smaller compared to the BreakHis dataset, the performance metrics were relatively lower. However, the experimental results demonstrates that the model still performed effectively, even with this limited dataset.Table 6 shows the performance of the model for BACH dataset, both for binary classification and multi classification task.

Also we have compared the performance of our model with few works from literature. Figure 10 shows the binary classification accuracy obtained for different approaches using BreakHis dataset.

To assess the significance of each component in our model, we conducted a series of ablation studies. Table 8 presents the results of experiments in which the SE residual convolutional blocks were replaced with standard convolutional layers for

TABLE 6. Classification performance of BACH dataset.

Classification	Precision	Recall	F1-Score	Accuracy
Binary	0.86	0.90	0.88	0.87
Multiclass	0.83	0.75	0.79	0.77

TABLE 7. Performance metrics on combined train set.

Tested on	Precision	Recall	F1-Score	Accuracy
BreakHis 200x	0.99	0.99	0.99	0.98
BreakHis200x&BACH	0.96	0.95	0.95	0.93

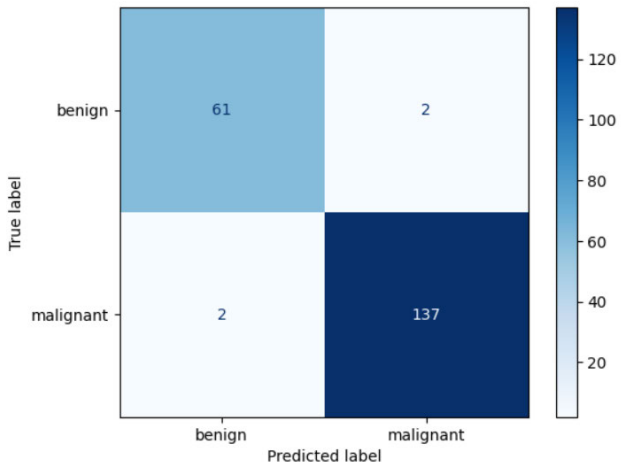

FIGURE 9. Confusion matrix for combined BreakHis-BACH sample images.

feature map generation, followed by a Conformer block. The findings indicate a significant decline in performance, despite an increase in the number of training epochs. This highlights the effectiveness of SE residual blocks in extracting relevant feature maps while simultaneously reducing computational complexity.

We further analyzed the impact of various architectural blocks to determine the optimal number of blocks that yield the best performance across different magnification levels. Table 9 presents a comparison of results on 200X magnification images by varying the number of SE Residual blocks, Conformer blocks, and attention heads.

To improve the model robustness and to enhance generalization, we combined the 200X magnification images from BreakHis and BACH datasets as the images are of same magnification.The performance evaluation was conducted by

TABLE 8. Results of 200x images with and without SE residual convolution blocks.

Initial Block	No of Conformer Blocks	Precision	Recall	F1-Score	Accuracy	No of Epochs
3 Convolution Layers	1	0.96	0.92	0.94	0.91	50
3 Convolution Layers	2	0.93	0.91	0.92	0.90	65
3 SE Res Conv Blocks	1	0.97	0.99	0.98	0.97	34
3 SE Res Conv Blocks	2	0.96	0.98	0.97	0.95	38

TABLE 9. Parameter sensitivity analysis of SE-conformer architecture with 200X images of BreakHis.

SE-ResConv Blocks	Attention Heads	Conformer Blocks	Precision	Recall	F1-Score	Accuracy	No of Epochs
2	4	1	0.96	0.95	0.95	0.93	35
3	4	1	0.98	0.98	0.98	0.97	75
3	2	1	0.97	0.99	0.98	0.97	34
3	2	2	0.96	0.98	0.97	0.95	38
4	4	1	0.93	0.96	0.94	0.92	33

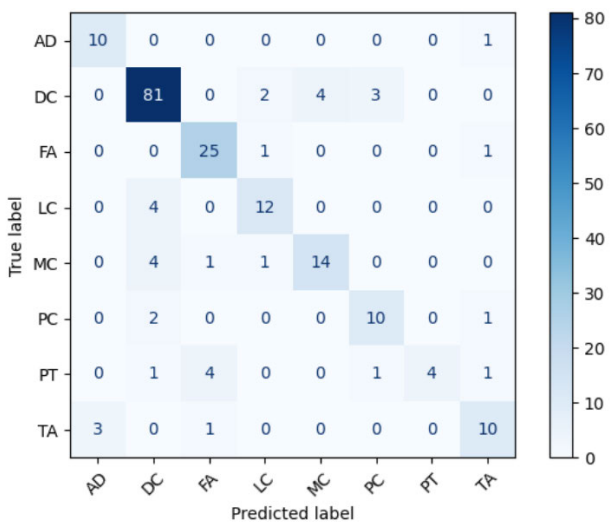

FIGURE 10. Confusion matrix for BreakHis multiclass classification.

TABLE 10. Performance comparison of SE-conformer model with state-of-the-art results.

References	Method used	Accuracy (%)
Rik Das et al. [9]	Feature generalization methods	88.4
Ping Wang et al. [10]	CNN + Capsulenet	94.52
BCHistoNet [11]	CNN architecture	89
S. Tariq et al. [35]	Vision Transformer	89.43
DecT [14]	Color deconvolution + Transformer	93.02
S. Boumaraf et al. [20]	Transfer Learning	94.49
Ting-Yan et al. [32]	Attention mechanism &	95.05
DRDA-Net [27]	Dual-shuffle attention guided DL model	97.43
Proposed Method	SE-convolutional blocks + Conformer	**98**

training on the combined datasets and testing exclusively on BreakHis images. The results, summarized in the table 7, demonstrates an improvement in performance compared to training with BreakHis images alone. This highlights the model's ability to achieve better results when provided with a larger and more diverse training dataset. The confusion matrix is shown in figure 9.Since we got best results for 200x magnification compared to the others, we have analyzed the model's performance by plotting the ROC curve (Receiver Operating Characteristic) for the classification of 200x images from BreakHis alone and also for the combined dataset, as shown in figure 8.It is evident that the model performs well giving an AUC(0.99) in both cases.

For a better estimate of the performance of our model, we used stratified K-Fold cross validation with (K = 5) on Breakhis 200x images and also on the combined dataset. The images were sourced from both datasets, along with approximately 500 augmented images per class. The results given in Table 11 shows that the model performs well without overfitting and is generalized to unseen data.

TABLE 11. Stratified 5-fold cross-validation results.

Fold	Loss	Accuracy	Precision	Recall	F1-score
1	0.324172	0.947291	0.948357	0.966507	0.957346
2	0.354177	0.938507	0.929224	0.973684	0.950935
3	0.341678	0.934114	0.942993	0.949761	0.946365
4	0.353778	0.942815	0.937644	0.971292	0.954172
5	0.317538	0.951613	0.948718	0.973684	0.961039
Average	**0.338269**	**0.942868**	**0.941387**	**0.966986**	**0.953971**

(a) Results for Combined dataset

Fold	Loss	Accuracy	Precision	Recall	F1-score
1	0.277285	0.973466	0.976316	0.981481	0.978892
2	0.311235	0.955224	0.965517	0.962963	0.964238
3	0.308624	0.965174	0.968504	0.976190	0.972332
4	0.240103	0.985050	0.986807	0.989418	0.988111
5	0.261205	0.978405	0.979003	0.986772	0.982872
Average	**0.279690**	**0.971464**	**0.975229**	**0.979365**	**0.977289**

(b) Results for Breakhis(200x) dataset

VI. CONCLUSION

In this paper, we introduced a novel deep learning architecture that integrates Squeeze-and-Excitation (SE) blocks within a convolutional-residual framework for effective feature extraction, followed by a Conformer block for further feature refinement before classification. The proposed model demonstrated superior performance, particularly in terms of F1-Score, when evaluated on the BreakHis and BACH datasets, while maintaining lower computational overhead compared to existing methods.

Furthermore, our model exhibited improved performance when trained on a combined dataset of BreakHis and BACH images, indicating its ability to generalize well with larger datasets and effectively handle diverse pathological variations. Various ablation studies were conducted to validate the effectiveness of the proposed approach, highlighting the importance of residual SE blocks in extracting relevant feature maps while reducing computational complexity. Additionally, the model's ability to achieve high accuracy within a shorter training duration reinforces its efficiency and suitability for real-world medical image classification tasks.

Our future work will focus on experimentation with real-time clinical data from hospitals to further validate and fine-tune the model.

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